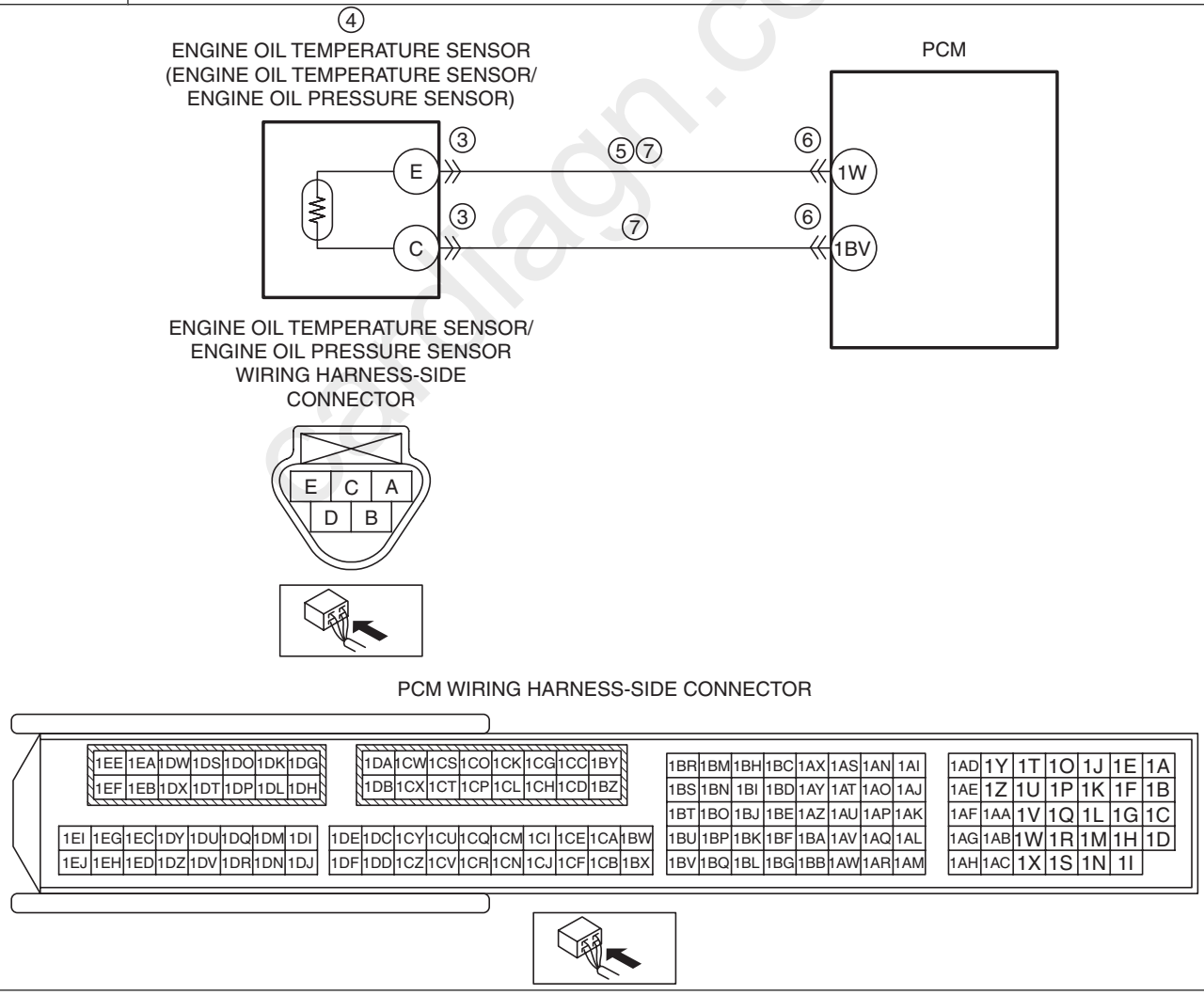
DTC P0197:00 [SKYACTIV-D 1.5]

id0102q2212000

DTC P0197:00	Engine oil temperature sensor circuit low input
DETECTION CONDITION	• The PCM monitors the input signal from the engine oil temperature sensor. If the voltage from the engine oil temperature sensor is **below 0.1 V** for **5 s**, the PCM determines that the engine oil temperature sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Engine oil temperature sensor/engine oil pressure sensor connector or terminals malfunction • Engine oil temperature sensor malfunction • Short to ground in wiring harness between engine oil temperature sensor/engine oil pressure sensor terminal E and PCM terminal 1W • PCM connector or terminals malfunction • Engine oil temperature sensor signal circuit and ground circuit are shorted to each other • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the engine oil temperature sensor/ engine oil pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
4	**DETERMINE IF ENGINE OIL TEMPERATURE SENSOR OR WIRING HARNESS MALFUNCTION** • Verify that the engine oil temperature sensor/ engine oil pressure sensor connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the engine oil temperature sensor/engine oil pressure sensor terminal E (wiring harness-side). • Is the voltage **approx. 5 V**?	Yes	Replace the engine oil temperature sensor/engine oil pressure sensor, then go to Step 8. (See ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**INSPECT ENGINE OIL TEMPERATURE SENSOR SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the engine oil temperature sensor/ engine oil pressure sensor connector is disconnected. • Switch the ignition off. • Inspect for continuity between engine oil temperature sensor/engine oil pressure sensor terminal E (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between engine oil temperature sensor/engine oil pressure sensor terminal E and PCM terminal 1W. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 8.
		No	Go to the next step.

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STEP	INSPECTION		ACTION
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
7	**INSPECT ENGINE OIL TEMPERATURE SENSOR SIGNAL CIRCUIT AND GROUND CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the engine oil temperature sensor/ engine oil pressure sensor and PCM connectors are disconnected. • Inspect for continuity between engine oil temperature sensor/engine oil pressure sensor terminals E and C (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Engine oil temperature sensor/engine oil pressure sensor terminal E—PCM terminal 1W • Engine oil temperature sensor/engine oil pressure sensor terminal C—PCM terminal 1BV **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to the next step.
		No	Go to the next step.
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.

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